



BIOLOGY CH.3 — NUTRITION IN PLANT — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway



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1. Respiration in Plants — Aerobic Process & Glucose Breakdown

- Aerobic respiration mein Oxygen zaroori hai — Pyruvate ka conversion CO_2 , H_2O aur energy mein O_2 ki presence mein hota hai. Glycolysis cytoplasm mein hoti hai, jisme glucose (6-carbon) oxidize hokar 2 pyruvate (3-carbon) molecules banata hai. Reaction: $\text{Glucose} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{Energy (ATP)}$. Anaerobic respiration mein oxygen ki absence mein incomplete oxidation hoti hai. [Q39]
- Glucose molecule breakdown hokar 'Pyruvic acid' ($\text{C}_3\text{H}_4\text{O}_3$) banata hai — Krebs cycle mein glucose ke oxidation se energy release hoti hai. Mitochondria cell ka respiratory organelle hai. [Q80]

2. Phloem — Structure, Companion Cells & Food Transport

- Phloem ek living vascular tissue hai jo leaves mein banaya gaya food (photosynthesis se) plant ke sabhi parts tak transport karta hai — is process ko 'phloem translocation' kehte hain. Phloem ke elements: sieve tubes, companion cells, phloem parenchyma, phloem fibres — sabhi bidirectional food transport karte hain. [Q40]
- Phloem diagram mein 'Companion Cell' hi perforated walls wala part NAHI hai (Sieve Pore perforated hoti hai) — Companion Cell sieve tube cells ki metabolic functioning mein help karti hai, transmembrane proteins use karke sugar-amino acids active transport se lete hai. [Q56]
- Phloem mein thickened cell walls wali cells milti hain — ye specialised food-conducting tissue hai, carbohydrates-amino acids ko leaf (photosynthesized part) se non-photosynthetic parts tak le jaati hai. [Q74]
- ⚠ Trap: Phloem transportation mein GTP aur CTP ka use NAHI hota — ye statement FALSE hai. Phloem transportation ATP energy use karta hai, sucrose active transport se load hoti hai, aur osmotic pressure badhne se water phloem mein move karta hai. [Q75]

3. Storage Carbohydrates — Starch (Plants) vs Glycogen (Animals)

- Plants mein storage carbohydrate 'Starch' hai — $(C_6H_{10}O_5)_n$, alpha D glucose units ka polymer. Glycogen animals-fungi mein glucose ki stored form hai, Sucrose (glucose+fructose) disaccharide hai, Glucose blood ka main sugar hai. [Q41]
- ⚠ Trap: Autotrophs unused carbohydrates ko 'Glycogen' ki tarah STORE NAHI karte — ye galat statement hai, wo Starch ki tarah store karte hain (Glycogen sirf animals mein hota hai). Autotrophs (green plants, kuch bacteria) 'primary producers' kehlaate hain — photosynthesis se carbon-energy requirements पूरी karte hain. [Q70]

4. Plastids — Types

- Colourless plastids 'Leucoplasts' kehlaate hain — endosperm, tubers, cotyledons, roots jaise non-photosynthetic parts mein milte hain. Chloroplasts light energy ko chemical energy mein convert karte hain (photosynthesis), Chromoplasts flowers-fruits ko yellow-orange-red color dete hain. [Q42]

5. Water's Role in Plant Cell Structure

- Plant cell mein 'Water' ki amount uski structure alter karke movement facilitate karti hai — plants eukaryotic cells hote hain. Plant hormones: Auxins, Gibberellins, Cytokinins, Ethylene, Abscissic Acid. Xylem water transportation facilitate karta hai. [Q43]

6. Iron-Rich Foods — Jaggery

- Jaggery mein iron high amount mein hota hai — 100 grams mein lagbhag 11 mg iron. Doosre iron-rich foods: nuts, dried fruit, legumes, dark leafy green vegetables (spinach, broccoli), oats. Milk-Butter Calcium-B2-B12 mein rich hain, Rice protein-vitamins ka source hai. [Q44]

7. Transportation in Plants — Xylem vs Phloem Features

- ⚠️ Trap: Water aur food ka movement 'bidirectional' NAHI hota (incorrect feature) — Xylem mein water movement unidirectional hota hai (roots se leaves), Phloem mein food bidirectional. Vessels-Tracheids dead cells hain, Sieve tubes living. Water movement physical forces se, food movement energy se hota hai. [Q45]
- Water transport ke baare mein sahi statement: 'Stomata ke through water vapour ki tarah loss hone ki process ko Transpiration kehte hain' — sahi hai. Xylem cells se leaves tak water pull hone ki process 'Translocation' NAHI, 'Ascent of sap' kehlaati hai. [Q63]

8. Why Plants Have Low Energy Needs

- Plants ki energy needs low hoti hain kyunki unke paas bahut se dead cells (jaise sclerenchyma) hote hain, jinhe koi energy nahi chahiye — animals ki tarah unhe food-shelter ki talaash mein move nahi karna padta. [Q46]

9. Autotrophic Nutrition — Essential Requirements

- ⚠️ Trap: 'Oxygen' autotrophic nutrition ke liye zaroori NAHI hai — Chlorophyll, Carbon dioxide aur Light zaroori hain. Autotrophic nutrition mein organism simple inorganic materials (water, mineral salts, CO₂) se apna food banata hai sunlight ki presence mein. [Q47]

10. Respiration & Gas Exchange in Plants

- ⚠️ Trap: 'Din mein atmosphere mein Carbon dioxide add hoti hai' ye statement INCORRECT hai — din mein plants oxygen release karte hain (photosynthesis), raat mein CO₂ release hoti hai (respiration). Plants stomata-lenticels se gases exchange karte hain, diffusion ke through. [Q48]
- Plant root cells oxygen 'soil particles ke beech ki air spaces' se lete hain — root hair mein oxygen diffuse hoti hai, respiration mein use hoti hai, CO₂ wapas usi rasta se escape hoti hai. [Q73]
- Transpiration 'Stomata' ke through hoti hai — leaves mein tiny openings/pores gas exchange allow karte hain. [Q101]
- Plants mein gases ka exchange 'Stomata' ke through hota hai — epidermis mein tiny pore, atmosphere aur plants ke beech gas exchange control karta hai (stomatal pore size change karke water loss bhi control karta hai). [Q102]

11. Plant Excretion — Waste Products

- ⚠️ Trap: Plant waste products 'centrioles' mein NAHI, 'cellular vacuoles' mein store hote hain — ye galat statement hai. Transpiration se surplus water nikalti hai, Photosynthesis oxygen by-product deti hai, Gum plants ka excretory product hai (old xylem mein resins-gums store hote hain). [Q49]

12. Translocation — Movement of Food

- 'Translocation' photosynthesis ke soluble products aur doosre materials ko leaves se plant ke doosre tissues tak transport karne ki process hai — Transpiration water evaporation hai, Guttation vascular plants (grass) se water droplets secretion, Tropic movement stimulus response movement hai. [Q50]
- Companion cells ki help se sieve tubes mein food-substances ka transport hota hai (upward-downward dono) — ise 'Translocation' kehte hain. [Q62]
- Spring season mein, roots mein stored nutrients 'Bud' ki taraf move karte hain — jab leaves shed hoti hain, roots mein stored sugar organic material ko growing Buds ki taraf mobilize karta hai — isse phloem mein bidirectional flow dikhta hai. [Q72]

13. Xylem — Structure & Components

- Xylem ke 'Tracheids aur Vessels' dead cells hote hain, phir bhi water-minerals transport karte hain — Xylem Fibre lignified dead cell hai (mechanical support deti hai), Xylem Parenchyma sirf living cells hain jo starch-fat store karte hain. [Q51]
- Xylem mein 'Parenchyma' cells food store karti hain — oil, fats, proteins, starch ke roop mein. Tracheid cells water-inorganic salts transport karte hain aur structural support dete hain. [Q54]
- ⚠️ Trap: 'Companion cell' Xylem ka component NAHI hai — Xylem ke components hain: Tracheids, Vessels, Xylem Fibre, Xylem Parenchyma (Companion cell Phloem ka part hai). [Q55]
- Xylem vessels 'Lignin' ki wajah se waterproof hote hain — Cellulose plant shape deta hai, Cutin cuticle mein waterproofing karta hai (leaves-fruits), Suberin cork cell walls mein water penetration rokta hai. [Q78]

14. Parasitic Nutrition — Cuscuta

- 'Cuscuta' parasitic nutritive strategy dikhati hai — heterotrophic nutrition ka ek mode, jisme parasite host ke body surface ya andar reh kar nutrition leta hai. Cuscuta, lice, leeches, tapeworms is category mein aate hain. [Q52]
- Cuscuta plants se parasite ki tarah nutrition leta hai — parasitic plants host plant se attach hokar special modified roots 'Haustorium' se nutrition lete hain. Doosre examples: Loranthus, Balanophora, Sandalwood tree. [Q104]

15. Photosynthesis — Carbon & Energy Requirements

- Plants ki carbon aur energy requirements 'Photosynthesis' se पूरी होती हैं — sunlight-water-CO₂ use करके oxygen aur sugar (energy) banate हैं. [Q53]
- Plants mein ultimate energy source 'Sunlight' hai — solar energy se producers (green plants) photosynthesis se food banate हैं, जो दूसरे organisms को transfer होता है। Photosynthesis rate Red colour mein maximum होती है. [Q83]
- Photosynthesis mein Oxygen 'release' होती है (atmosphere mein) — green plants CO₂-water se carbohydrates banate हैं, sunlight की presence mein, chemical energy release करते हैं. [Q98]
- ⚠️ Trap: 'Oxygen' photosynthesis के लिए essential NAHI है — Chlorophyll, Sunlight aur Carbon dioxide zaroori हैं, Oxygen एक by-product है जो release होती है. [Q99]
- Photosynthesis mein plants 'Solar energy' (light energy) use करते हैं — chlorophyll जैसे pigments इसे capture करके chemical energy (glucose) mein convert करते हैं. [Q100]

15b. Oxygen — Waste Product of Photosynthesis

- Photosynthesis के दौरान 'Oxygen' waste product की तरह produce होती है — CO₂ aur water, sunlight-chlorophyll की presence mein, oxygen aur glucose mein convert होते हैं। Glucose plant food की तरह use होता है, oxygen by-product है: $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$. [Q57]

16. Atmospheric Pressure & the Rate of Transpiration

- Atmospheric pressure aur transpiration rate 'inversely proportional' होते हैं — low pressure mein water vapour fast move करता है (transpiration बढ़ती है), high pressure mein water loss slow होता है. [Q58]

17. Testing for Starch in Photosynthesis Experiments

- Photosynthesis experiments mein starch की presence measure करते हैं क्योंकि 'Glucose formed starch mein convert हो जाता है' — Glucose plant cell growth (cellulose) के लिए use हो सकता है, starch mein convert होकर बाद mein wapas glucose बन सकता है, या respiration mein energy release कर सकता है. [Q59]

18. Active Transport — Food, Amino Acids & Hormones

- Plants mein diffusion से ज्यादातर transportation होती है, लेकिन कुछ substances (Food, Amino acids, Hormones) के transport के लिए energy चाहिए — ये 'Active Transport' कहलाता है। Passive transport (simple diffusion) mein energy expenditure नहीं होती. [Q60]

19. Plant Nutrition — Mineral Absorption

- ⚠️ Trap: 'Minerals-nutrients water se poorly absorb hote hain' — ye statement INCORRECT hai (galat) — actually minerals-nutrients water ke through hi absorb hote hain (root zone soil ke water se). Main nutrients (NPK): Nitrogen-Phosphorus-Potassium, aur Calcium-Magnesium-Sulphur. Iron-Manganese micronutrients hain. [Q61]
- Nitrogen plants ko soil se milta hai — plants khud atmosphere se nitrogen absorb nahi kar sakte, ammonia ya nitrates ke roop mein (nitrogen fixation se) rely karte hain. Main nutrients: Nitrogen, Phosphorus, Potassium. [Q89]

20. Photosynthesis — Process, Sequence & Products

- ⚠️ Trap: 'Photosynthesis mein CO₂ ka carbohydrates mein oxidation hota hai' ye statement INCORRECT hai (galat) — actually CO₂ ka 'reduction' hota hai carbohydrates mein, na ki oxidation. Equation: $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$. [Q64]
- Photosynthesis ke liye 4 components zaroori hain: Carbon dioxide, water, sunlight aur chlorophyll — oxygen by-product ki tarah release hoti hai, food carbohydrates (glucose-starch) ki tarah store hota hai. [Q65]
- Photosynthesis ke steps ka sahi sequence: (c) chlorophyll dwara light absorption → (b) light energy chemical energy mein convert → (a) water molecule split (H aur O mein) → (d) CO₂ reduce hokar carbohydrates. [Q66]
- Photosynthesis mein 6 water molecules '12H₂O' se produce hote hain — glucose banane ke liye 12 water molecules aur 6 CO₂ chahiye: $6\text{CO}_2 + 12\text{H}_2\text{O} + \text{Light Energy} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 + 6\text{H}_2\text{O}$. [Q69]
- ⚠️ Trap: 'Oxygen ka absorption' photosynthesis ke dauran NAHI hota — light energy absorption, water molecules ka splitting, aur CO₂ ka reduction hota hai, oxygen absorb nahi, release hoti hai. [Q77]
- Green light range photosynthesis ki rate ko sabse kam affect karta hai — chlorophyll green wavelength absorb nahi karta (isliye plants green dikhte hain), red-blue light zyada effective hoti hai photosynthesis ke liye. [Q85]
- Photosynthesis se 'Glucose, Water aur Oxygen' products banate hain — $6\text{CO}_2 + 12\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6$ (Glucose) + $6\text{H}_2\text{O}$ (Water) + 6O_2 (Oxygen). [Q105]

21. Plant Hormones — Cytokinin & Fruits

- 'Cytokinin' hormone fruits mein particularly high hoti hai — cell division promote karti hai aur cell expansion badhati hai. Gibberellin germination-elongation control karti hai, Auxins embryonic development-tropisms, Absciscic acid growth-stress response regulate karti hai. [Q67]

22. Saprotrophic Nutrition — Mushroom

- Mushrooms mein 'Saprotrophic' nutrition hoti hai — organisms dead-decaying organic matter se food lete hain. Doosre saprotrophic organisms: Rhizopus (bread mould), Mucor (pin mould), Yeast. [Q68]
- Mushroom ek 'Saprotroph' hai — dead-decaying matter se solution form mein food leta hai. Doosre saprotrophs: Bacteria, fungi, molds, Mycorrhizal fungi. Algae autotrophs hain, Pigeon-Man omnivorous hain. [Q103]

23. Xylem-Phloem Transport — Direction

- Diagram mein blue arrows (Xylem transport) aur red arrows (Phloem transport) — Xylem: root hair se cell-to-cell osmosis se xylem tak, phir leaves tak. Phloem: leaves se nutrients-sugar ka bidirectional translocation. [Q71]

24. Water Transport Pathway — Soil to Leaves

- Water ka soil se leaves tak transport pathway: Soil → root hairs → root cortex → xylem of roots → xylem of stem → xylem of leaf — root hair diffusion se water-minerals soil se absorb karte hain, xylem small vessels roots-leaves ko connect karte hain. [Q76]

25. Fibrous Roots — Rice, Wheat, Maize (NOT Beetroot)

- ⚠️ Trap: Beetroot mein fibrous roots NAHI hoti (taproot hoti hai) — Rice, Wheat, Maize jaise plants mein fibrous roots hoti hain (thin, moderately branching, dense network, soil surface ke close). Doosre examples: marigold, Sugarcane, banana. [Q79]

26. Sugarcane — Efficient Converter of Sunlight

- Sugarcane plant sunlight ko chemical energy mein sabse efficiently convert karta hai — photosynthesis se sunlight-CO₂-water use karke glucose-oxygen banata hai. Sugarcane (*Saccharum officinarum*) Poaceae family ki tall perennial grass hai. [Q81]

27. Phototropism — Sunflower Facing the Sun

- Sunflower ka sun ki taraf move karna 'Phototropism' kehlaata hai — plant light ki direction mein grow karne ki ability, jisse photosynthesis ke liye sunlight exposure maximize hoti hai. Charles Darwin ne 1880 mein isse study kiya, 'The Power of Movement in Plants' book mein publish kiya. [Q82]

28. Chlorophyll — The Green Pigment

- 'Chlorophyll' plants ko green color deta hai — green wavelength ko absorb nahi karta, isliye plants green dikhte hain. Chloroplasts mein milta hai, photosynthesis ke liye responsible, aur Magnesium element contain karta hai. [Q84]

29. Desert Plants — Photosynthesis in the Stem

- Cactus jaise desert plants (Xerophytes) mein photosynthesis 'Stem' mein hoti hai — deep-spreading roots aur water-storing capacity ke saath dry conditions withstand karte hain. Examples: Aeonium, Agave. [Q86]

30. Vitamin C — Gooseberry

- Gooseberry (Amla) Vitamin C (Ascorbic acid) ka rich source hai — doosre sources: Citrus fruits (Oranges, Kiwi, Lemon, Grapefruit), Strawberries, Peppers. [Q87]

31. Rhizobium — Root Nodules of Leguminous Plants

- 'Rhizobium' bacteria leguminous plants ke root nodules ke formation se associate hai — nitrogen fixation mein help karta hai. (Salmonella, Clostridium, E.coli inse alag hain.) [Q88]

32. Vitamin B — Cereals Are the Richest Source

- Cereals Vitamin B ka richest source hain. Doosre Vitamin sources: Vitamin A (Carrots, tomatoes, cod liver oil), Vitamin C (Orange, amla, lemon), Vitamin D (Fish oil, sunlight, milk), Vitamin E (Spinach, nuts). [Q90]

33. Osmosis — Water Uptake by Root Hairs

- Soil se water root hairs mein 'Osmosis' ki wajah se enter karta hai — root hairs delicate epidermal outgrowths hain, large surface area rakhte hain, semi-permeable membrane se solvent molecules concentration gradient ke against move karte hain. [Q91]

34. Nitrogen Fixation — Legumes

- 'Legumes' apne aap air se nitrogen fix kar sakte hain (Fabaceae family) — root nodules mein Rhizobium bacteria ki help se. Tubers doosri category ke plants hain (storage organs). [Q92]

35. Dietary Fiber — Cellulose (Insoluble)

- 'Cellulose' insoluble dietary fiber ka example hai — seed-skin of fruit, whole-wheat, bread, brown rice mein milta hai. Soluble fibers (Psyllium, Pectin, Inulin) digestion slow karte hain, fatty acids ko bind karte hain. [Q93]

36. Banana — Rich in Potassium

- Banana mein sabse zyada amount mein 'Potassium' hota hai — carbohydrates, fibers, Vitamin B6, Manganese bhi hote hain. Doosre Potassium sources: Potatoes, Spinach, Avocados. [Q94]

37. Plasmolysis — Water Loss by Osmosis

- 'Plasmolysis' living plant cell ka osmosis se water lose karke shrink hone ki process hai — protoplasm shrink hota hai. Cytolysis (excess water se cell burst hona) aur Hemolysis (RBC breakdown) isse alag hain. [Q95]

38. Minerals in Diet — Fluoride & Potassium

- Coffee, spinach, onion aur tea 'Fluoride' provide karte hain diet mein — tooth enamel remineralize karta hai. (Sodium — canned meat, bacon; Iodine — fish, seaweed; Chloride — olives, tomatoes.) [Q96]
- Green aur yellow vegetables diet mein 'Potassium' provide karte hain. (Zinc — sea foods, eggs; Sodium — apples, guavas; Copper — meats, shellfish.) [Q97]

39. Meristems — Primary Growth (Intercalary & Apical)

- Plants mein primary growth 'Intercalary aur Apical meristem' se hoti hai (C and D) — Apical meristem root-shoot tip par growth karti hai, Intercalary meristem internodes/leaf base par length badhati hai. Lateral meristem thickness badhati hai (secondary growth). [Q106]